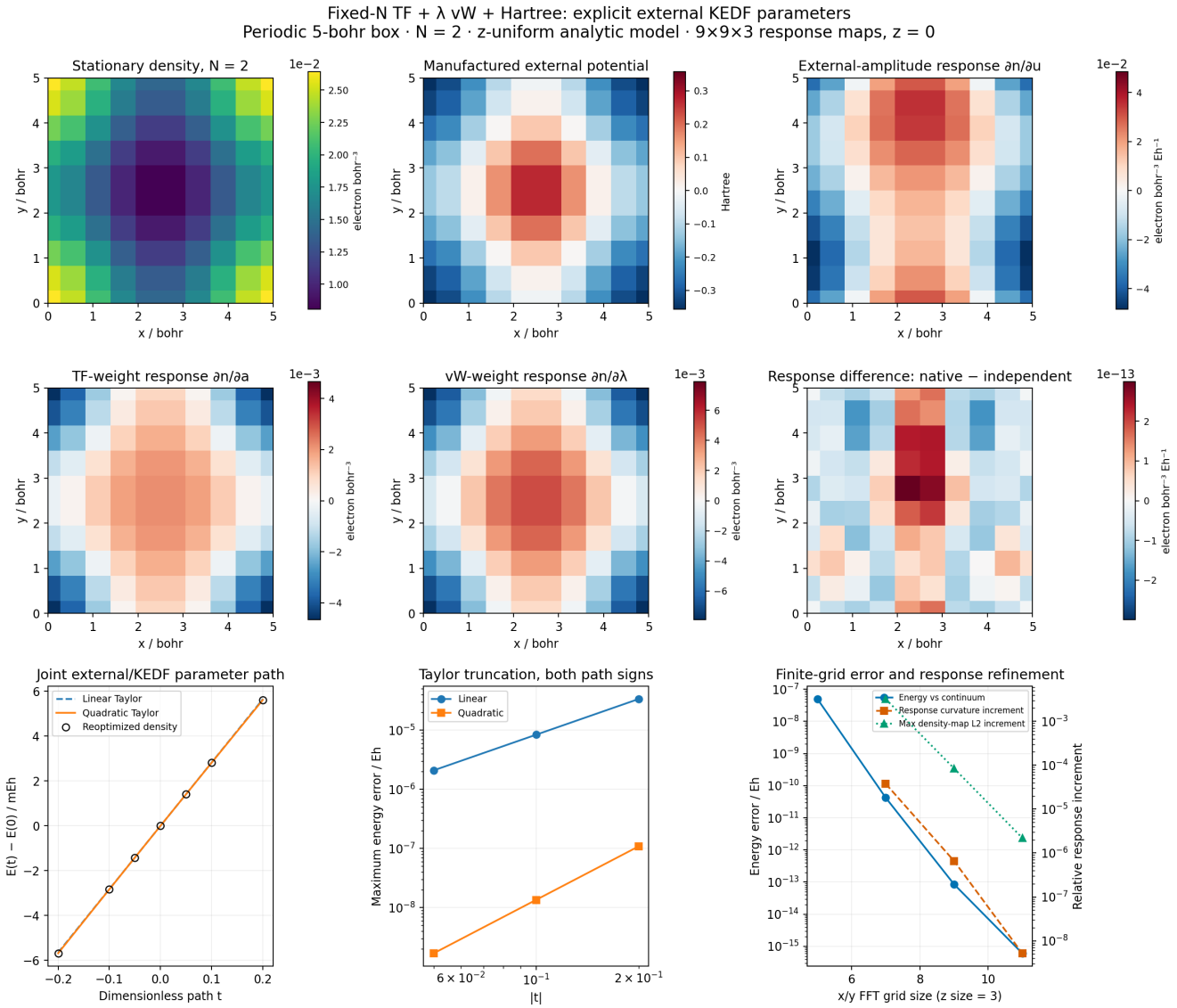
P08 — fixed-electron OFDFT parameter and density response

This reproducible **analytic periodic model** verifies an external TF/vW kinetic functional, density response, and normalized second-order Taylor coefficients using the existing OFDFT optimizer and DFPT solvers. It contains two electrons in a 5-bohr cubic cell, a manufactured smooth local potential and a neutralizing Hartree background. There are no atoms, pseudopotentials or XC. It is not a material benchmark or evidence of the chemical accuracy of TF/vW.



Native results · independent analytic checks · density/response CSV · PDF · ordinary calculation script · independent comparison · plot

Functional, constraint and external parameter interface

In atomic units the declared model is

```
E[n] = a C_TF integral n^(5/3) + (lambda/8) integral |grad n|²/n
       + integral v_ext n + (1/2) integral n v_H[n]
C_TF = (3/10) (3 pi²)^(2/3)
integral n = N = 2
```

With the real amplitude `phi=sqrt(n)`, the vW term is `(lambda/2) integral |grad phi|²` and its potential is `−lambda Delta phi/(2 phi)`. These are the standard spin-unpolarized TF and vW conventions; the latter potential is also directly inspectable in DFTpy's implementation.
The independent calculation here uses mathematical formulas and NumPy/SciPy, not a DFTpy optimization run.

The Hartree kernel is `4 pi/G²` at nonzero G and **zero at G=0**. The fixed uniform compensating background removes that electrostatic zero mode. This convention does not discard a constant external potential: at fixed N, adding `t` to the potential changes the chemical potential by `t` and the energy by `N t`, while leaving the density unchanged.

`external_kinetic(parameters, features)` is an ordinary external callback that delegates the two kinetic terms to `ofdft.KineticFunctional('tfvw')`. Its nested `parameters['weights']['tf'/'vw']` are explicit PyTree leaves. No traced trainable weight is captured in a closure. `run_ofdft` relaxes the density for each parameter value; `dfpt.ofdft_problem` supplies constrained linear response and Taylor propagation. No optimizer, response solver or KEDF implementation was added to `src`.

A manufactured nonuniform density with a known chemical potential

Let `k=2 pi/5`, `b=0.18`, `c=0.12`, and

```
phi = A (1 + b cos(kx) + c cos(ky))
A² = (N/V) / [1 + (b²+c²)/2]
n_star = phi²
Delta phi = −A k² [b cos(kx)+c cos(ky)]
v_ext = mu_star − (5/3) a C_TF n_star^(2/3)
        + (lambda/2) Delta phi/phi − v_H[n_star]
```

The baseline coefficients are `a=0.8`, `lambda=0.6`, and `mu_star=0.17 Eh`. The amplitude is strictly positive. Its Fourier support ends at index one, and the density/Hartree support at index two, so odd x/y meshes of size at least five represent those fields without aliasing. The

calculation starts from a uniform amplitude rather than the manufactured solution: the four solves require 56, 117, 131 and 101 iterations.

Once the external potential has been constructed, it is **held fixed when kinetic parameters change**. Rebuilding the manufactured potential with the changed weights would instead cancel the intended density response. The perturbed potential is `v_ext+u p(r)`, with `p=cos(kx)+0.4 sin(ky)`, and `u` in Hartree. The other two parameters are dimensionless. Density units are electron/bohr³; its derivatives with respect to u additionally carry `Eh^-1`.

The model is uniform in z. The grids are `(5,5,3)`, `(7,7,3)`, `(9,9,3)` and `(11,11,3)`; only x/y cutoffs are refined. The main maps use `(9,9,3)` and show nearest sampled values at z=0, with periodic boundary samples duplicated for display. No visual smoothing is interpreted as numerical refinement.

Independent fixed-N response

For a homogeneous density `n0=N/V` and nonzero Fourier wavevector G, the analytic response to a potential component is

```
delta n_G = −delta v_G /
    [(10/9) a C_TF n0^(−1/3) + lambda |G|²/(4 n0) + 4 pi/|G|²].
```

Three focused cases check TF-only, vW-only and TF+vW **kinetic terms**, with Hartree retained in all cases. The constrained zero Fourier component is zero; no division by G=0 is used. Constant-potential response separately checks `delta n=0`, `delta mu=1` and `delta E=N`.

For the nonuniform reference, the independent code constructs dense Fourier Laplacian `Delta` and Coulomb `H` matrices using NumPy, then forms the functional-derivative Hessian

```
K = diag[(10/9) a C_TF n^(−1/3)] + H
    − (lambda/4) diag(1/phi) Delta diag(1/phi)
    + (lambda/4) diag(Delta phi/phi³).
```

SciPy solves the bordered system `[K,−1; 1.T,0] [delta n,delta mu] = [−source,0]`. Uniform voxel weights make `1.T delta n=0` equivalent to fixed integrated N. The sources are p for u, `(5/3) C_TF n^(2/3)` for a, and `−Delta phi/(2 phi)` for lambda. This comparison imports no GradSCF physics, AD derivatives or optimizer.

At the main grid, the complete three-column implicit density Jacobian agrees with this independent KKT result within **5.73e-13**. The implicit/DFPT Jacobians agree within `1.18e−16`, and the density-observable JVP/VJP adjoint identity within `1.78e−15`. The derivative of integrated N is checked for every parameter. Across all grids, the manufactured density differs by at most `9.68e−13 electron/bohr³`, the electron count by `3.2e−15`, and the chemical potential by `9.3e−13 Eh`.

Taylor convention and finite differences

The joint dimensionless path is

```
(u/Eh, a, lambda)(t) = (0, 0.8, 0.6) + t (0.03, 0.05, −0.04).
E(t) = E0 + c1 t + c2 t² + ...
c1 = 0.0282497967358199 Eh
c2 = −0.000835141384583721 Eh = E''(0)/2!
```

Density coefficients use the same `derivative/k!` convention. Both stored density orders integrate to zero. For this affine parameter path, the independent response gives `E''=dtheta.T source.T W density_jacobian dtheta`; its half reproduces c2. Both energy coefficients agree with the independent result within `1.05e−16 Eh`. The independently assembled energy parameter Hessian is symmetric and negative semidefinite, as expected for minimizing an energy affine in these parameters under a fixed constraint.

Every parameter is checked with central steps `1e−3`, `3e−4`, `1e−4`, in its declared units. The finest maximum density-response and energy-gradient errors are `4.95e−9` and `1.70e−9`. Joint-path second coefficients use steps `0.02`, `0.01`, `0.005`; the finest density/energy coefficient errors are `2.33e−8` and `3.76e−11`. Density second differences stop improving at the smallest steps, consistent with the finite forward accuracy divided by step squared. The complete nonmonotone sequence is retained and is not presented as an asymptotic truncation-error estimate.

At `|t|=0.2`, the worst quadratic energy prediction error is about `1.08e−7 Eh`, versus `3.36e−5 Eh` for the linear approximation. Halving |t| gives approximately the expected cubic versus quadratic truncation trend. These local expansions do not establish reliability across density nodes or changes in the solution branch.

Discrete agreement versus grid error

The analytic manufactured density is stationary on each selected grid, but the nonpolynomial TF energy and the response still have discretization error. An independent 2D Gauss–Legendre integral of the known smooth density provides

the continuum TF energy; the Hartree energy is a finite analytic Fourier sum. Orders 48 and 80 agree within `9.16e-16 Eh`. The manufactured identity is `E=mu_star*N-(2/3) a T_TF-E_H`; the vW terms cancel only because this particular external potential was constructed from the baseline solution.

x/y size; z size=3	Energy / Eh	Difference from continuum quadrature / Eh	Potential-response curvature / Eh^-1
5	0.124417330510597	4.92e-8	−2.092717042641
7	0.124417281338339	4.19e-11	−2.092638431423
9	0.124417281296526	8.59e-14	−2.092637050669
11	0.124417281296441	6.11e-16	−2.092637039796

The last energy difference is at the quadrature/float64 floor; it is not a general accuracy claim. The last scalar-curvature increment is `5.20e-9` relative. Separately, Parseval norms compare the **entire Fourier-interpolated density responses** on successive grids. For potential/a/lambda, the last relative L2 changes are `1.21e-6`, `6.83e-7` and `2.21e-6`, all below the `2e-5` increment gate. These are refinement increments, not rigorous bounds on unknown continuum-response errors. The `5.73e-13` KKT agreement is strictly a same-grid implementation check and must not be substituted for these grid errors.

Validation and reproduction

Five new focused tests passed in **71.05 s**, following five expected failures before the new example entrypoints existed. The existing kinetic/OFDFT/DFPT response regressions passed: **17 passed in 283.58 s**. No tests were skipped. The four-grid native run took **151.52 s**, peak RSS **3,040,204 KiB**, on `jy_gpu_1`/c20, Intel Xeon Platinum 8457C, Python 3.11.15, JAX 0.10.0, CPU float64. Numerical calculations and plotting were remote; local checks were static syntax/hash/link validation and visual QA.

The independent snapshot is `/home/yjiao/gradscf_checks/ad-showcase-p08-20261016`, cloned from the frozen integration source, with no OFDFT/core modifications. Its Python source SHA256 is `c792320a88f24c695049469ac317d26953efc0888cd3236c44cb0fb393220abc`.

```
export CUDA_VISIBLE_DEVICES= JAX_PLATFORMS=cpu JAX_ENABLE_X64=1
export JAX_ENABLE_COMPILATION_CACHE=false OMP_NUM_THREADS=2
OPENBLAS_NUM_THREADS=2
export PYTHONPATH=src
PYTHON=/home/yjiao/opt/miniconda3/envs/jax_scf/bin/python
timeout 900 "$PYTHON" -m pytest -q tests/ofdft/test_showcase_response.py
timeout 1800 "$PYTHON" examples/ofdft/parameter_density_response.py
timeout 300 "$PYTHON"
```

```
examples/ofdft/compare_parameter_density_response.py
timeout 300 "$PYTHON" examples/ofdft/plot_parameter_density_response.py
```

The four raw mesh NPZ files and independent response NPZ are included in this compact result bundle. Source manifest, verification manifest, red/green/regression logs and the two calculation records preserve the executed source, parameters, thresholds and evidence. No empirical fit, trained functional, atomistic accuracy, physical crystal response or material convergence is implied.